Jiwon Kang

Last update: October 30, 2026

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Research Interests

Having started my research career in 3D vision as an undergraduate researcher, I expanded my focus during my M.S. to Visual Generative AI and Unified Multimodal Models (UMMs), leading to four (co-)first-author papers at NeurIPS, ICLR, CVPR, and ECCV. My work primarily spans diffusion-based visual generation and the interaction between multimodal understanding and generation. Building on my experience, my current research interests extend toward video diffusion or world (action) models, as well as multimodal generative models that bridge perception and generation across images, video, audio, and action.

Key research areas include:

- Diffusion & Flow-based Generative AI (Image/Audio Generation, Guidance, Sampling) [NeurIPS'25, ICLR'26, CVPR'26, Kakao]
- Multimodal Understanding & Generation (VLMs, UMMs) [ECCV'26]
- 3D Vision (Novel View Synthesis, NeRF, Gaussian Splatting) [ICCVW'25, WACV'27]

Availability: Available for full-time roles after completing my M.S. in February 2027. Open to internship or other pre-graduation opportunities before then.

Education

M.S. in Artificial Intelligence — Mar. 2025 – Feb. 2027 (Expected)
Kim Jaechul Graduate School of AI, KAIST — Advisor: Seungryong Kim (CVLAB)

B.S. in Computer Science and Engineering — Mar. 2019 – Feb. 2025
Korea University

- GPA: 4.24 / 4.5
- Double Major in Statistics
- Leave of Absence for R.O.K. Military Service (Dec. 2020 – Jun. 2022)
- Undergraduate Research Intern @ CVLAB (Sep. 2022 – Feb. 2025), Advisor: Seungryong Kim

Experience

Kakao — Mar. 2026 – Jun. 2026
Research Intern — *Seoul, Korea*

- Developed a Whisper-based continuous speech tokenizer and an autoregressive diffusion TTS system, adapting diffusion modeling to high-dimensional speech representations [P1].

Projects

SAMSUNG — Aug. 2025 – Oct. 2025
Research Collaboration — *Seoul, Korea*

- Developed a diffusion-based attribute-preserving face-swapping framework with Samsung Electronics for background face anonymization [C3].

Publications

*: Equal Contribution, †: Corresponding Author

International Conference

[C5] RAIN-GS: Relaxing Accurate Initialization Constraint for 3D Gaussian Splatting Paper | Project Page | Code

Jaewoo Jung*, Jisang Han*, Honggyu An*, ***Jiwon Kang***, Seonghoon Park*, Seungryong Kim†

- Analyzed why 3D Gaussian Splatting critically depends on accurate SfM initialization, showing that limited Gaussian transport and missing low-frequency scene structure prevent robust coarse-to-fine optimization from random or noisy point clouds.
- Proposed *RAIN-GS* with Sparse-Large-Variance initialization, progressive Gaussian low-pass filtering, and Adaptive Bound-Expanding Split, enabling 3DGS to train from sub-optimal or random point clouds with performance comparable to or better than SfM initialization.

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027).

[C4] Transferability Between Understanding and Generation in Unified Multimodal Models Paper | Project Page | Code

Jiwon Kang, Heeji Yoon*, Jaewoo Jung, Jaewon Min, Minkyeong Jeon, Biyeon Hwang, Sangwon Jung†, Seungryong Kim†

- Investigated cross-task transferability between image understanding and generation in Unified Multimodal Models, revealing that its strength strongly depends on architecture and emerges most clearly with a fully shared Transformer backbone and unified visual encoder.
- Leveraged understanding-to-generation transfer as a practical post-training strategy to improve targeted generative capabilities, including counting, spatial relations, and text rendering, while mitigating the distribution shift and quality degradation caused by direct generation fine-tuning.

European Conference on Computer Vision (ECCV 2026).

[C3] APPLE: Attribute-Preserving Pseudo-Labeling for Diffusion-Based Face Swapping Paper | Project Page | Code

Jiwon Kang, Yeji Choi*, JoungBin Lee, Wooseok Jang, Jinhyeok Choi, Taekeun Kang, Yongjae Park, Myungin Kim, Seungryong Kim†

- Identified poor attribute preservation in diffusion-based face swapping caused by masked inpainting formulations, and designed an attribute-preserving teacher using (1) conditional deblurring and (2) attribute-aware inversion to retain both global and fine-grained target attributes.
- Constructed high-fidelity pseudo-triplets with the teacher and used them to train a direct-editing student under clean attribute conditions. The resulting model achieved state-of-the-art attribute preservation with competitive identity transfer, consistently performing on or near the Pareto frontier across datasets.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026).

[C2] A Noise is Worth Diffusion Guidance Paper | Project Page | Code

Donghoon Ahn*, ***Jiwon Kang***, Sanghyun Lee, Jaewon Min, Wooseok Jang, Minjae Kim, Hyungwon Cho, Sayak Paul, SeonHwa Kim, Eunju Cha†, Kyong Hwan Jin†, Seungryong Kim†

- Revealed that high-quality guidance-free generation is largely determined by the initial noise, showing that low-frequency components transform Gaussian noise into a favorable noise space for unguided denoising.
- Proposed *NoiseRefine*, which learns to refine the initial noise once before sampling, reproducing guidance-level generation quality without additional guidance passes while reducing inference cost and requiring only 50K text-image pairs for training.

International Conference on Learning Representations (ICLR 2026).

[C1] Where and How to Perturb: On the Design of Perturbation Guidance in Diffusion and Flow Models Paper | Project Page | Code

Donghoon Ahn*, ***Jiwon Kang****, Sanghyun Lee, Minjae Kim, Wooseok Jang, Jaewon Min, Sangwu Lee, Sayak Paul, Seungryong Kim†

- Conducted the first head-level analysis of attention perturbation guidance in Diffusion Transformers, revealing that individual attention heads exhibit specialized and compositional effects on visual concepts such as structure, lighting, style, and texture.
- Proposed *HeadHunter* to iteratively select attention heads for arbitrary user-defined objectives and *SoftPAG* to continuously control perturbation strength, enabling fine-grained quality and style guidance on SD3 and FLUX.1 with compact head subsets and negligible inference overhead.

Conference on Neural Information Processing Systems (NeurIPS 2025).

Workshop

[W1] Self-Evolving Neural Radiance Fields Paper | Project Page | Code

Jaewoo Jung*, Jisang Han*, ***Jiwon Kang****, Seongchan Kim, Min-Seop Kwak, Seungryong Kim†

- Formulated few-shot NeRF as an iterative teacher-student self-training problem, progressively expanding supervision with pseudo-labeled novel views generated by the teacher and promoting each trained student to the next teacher.
- Introduced cross-view semantic consistency based ray reliability estimation and reliability-aware distillation, allowing existing NeRF models to learn more robust geometry from sparse views without external 3D priors or architectural modifications.

ICCV 2025 Workshop on Wild3D: 3D Modeling, Reconstruction, and Generation in the Wild.

Preprints & Under Review

[P1] Exploring Whisper as a Continuous Tokenizer for Text-to-Speech

Jiwon Kang, Heeji Yoon, Woo-Young Kang, Sunghun Kang, Jihwan Eom, Jaeyeol Jeon, Byungseok Roh, Seungryong Kim†

- Inspired by the success of RAE in the vision domain, explored whether ASR-supervised audio representations (Whisper) could be repurposed as generative representations for TTS.
- Demonstrated that frozen Whisper features support feed-forward, variable-length, multi-speaker reconstruction, showing that ASR-supervised semantic representations retain acoustic information for direct waveform reconstruction.
- Built an autoregressive diffusion-based TTS system that directly models Whisper features and showed that RAE-inspired adaptations for high-dimensional representations improve their training efficiency, enabling intelligible speech generation.

Under Review.

Honors

Outstanding Paper Silver Award, *IPIU 2024* 2024

- **Paper:** *Self-Evolving Neural Radiance Fields*

Extracurricular Activities

Military Service (KATUSA, Korean Augmentation to the U.S. Army) Dec. 2020 – Jun. 2022

- MOS : 92Y (Unit Supply Specialist)
- 194th DSSB, 2nd Infantry Division, Eighth U.S. Army, Camp Humphreys

AIKU (Artificial Intelligence Society of Korea University) Sep. 2022 – Aug. 2023

- Co-founder & Vice President of academic research society for undergraduate students